

Problem Statement Catalog

1 Statement Exported

[PS-011]

NOT SPECIFIED IMPACT

SIH 2026

3D ULPIN Generation and vertical Property Mapping System

ORGANIZATION / MINISTRY:

Ministry of Rural Development

CATEGORY: Software

THEME: Space Technology

GEOGRAPHY: India

DOMAINS: Software, Space Technology

DEMOGRAPHICS: General Public / All

TIMELINE: SIH 2026

BACKGROUND

With rapid urbanization and vertical growth of cities, conventional 2D land record Systems are becoming inadequate for managing modern urban properties. Existing land administration systems are primarily designed to identify surface-level land parcels and are unable to uniquely define ownership rights associated with multi

- storey apartments, underground infrastructure, elevated transport corridors, parking spaces, air-rights, and subsurface utility networks.

DESCRIPTION

The proposed solution Should develop an advanced 3D ULPIN(Unique Land Parcel Identification Number) Generation and vertical Property Mapping system capable of creating unique spatial identities for: . Surface land parcels . Multi-storey apartments . Underground infrastructure The system should integrate: . Drone imagery . LiDAR/3D Point cloud data . GIS parcel layers . Building floor Plans . GNSS/CORS-based coordinates . Digital Elevation Models (DEM/DSM) The solution should also incorporate AI/ML capabilities for: . Automated building extraction . Floor segmentation . Vertical Parcel delineation . Intelligent topology validation

EXPECTED SOLUTION

The expected outcome is development of a scalable and interoperable 3D cadastral framework capable of: . Generating standardized 3D ULPINs . Mapping vertical and underground ownership rights . Supporting volumetric cadastre systems . Enabling accurate urban property governance . Reducing ownership conflicts and ambiguities . Improving infrastructure planning and utility management

TAGS:

#ulpin #generation #vertical #property #mapping #space

SOURCES:

<https://sih.gov.in/sih2026PS>